Determining the Effect of the Combined Treatment of Sulforaphane and Luteolin as an Inhibitor of Extracellular Regulated Kinases in 4T1 Cells

Introduction

Figure 1: An image of a breast cancer statistic https://www.nationalbreastcancer.org/breast-cancer-facts/

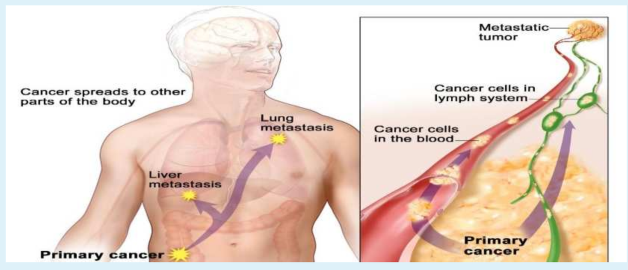


Figure 2: An image showing the development of cells that are metastatic https://www.cancer.gov/publications/dictionaries/cancer-terms/def/metastasis

Metastasis:

- The uncontrolled growth and spread of cancer cells to other parts of the body
- 90% of breast cancer related deaths

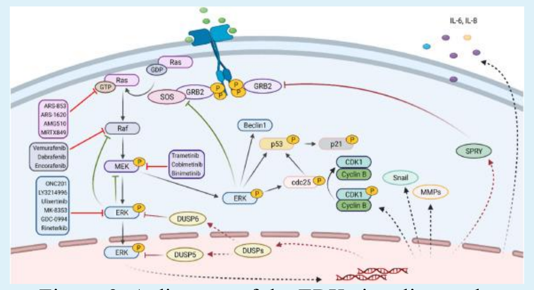


Figure 3: A diagram of the ERK signaling pathway https://www.sciencedirect.com/science/article/pii/S2352304222001404

Extracellular Regulated Kinase (ERK)

- Signaling pathway in healthy and abnormal cells
- Regulates cell proliferation and causes metastasis

Treatments

Figure 4: The chemical structure of sulforaphane https://www.medchemexpress.com/r-sulforaphane.html

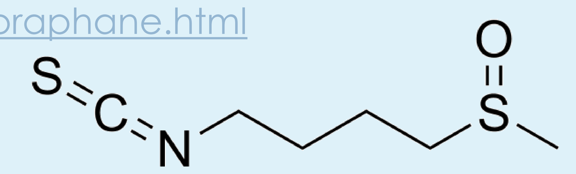
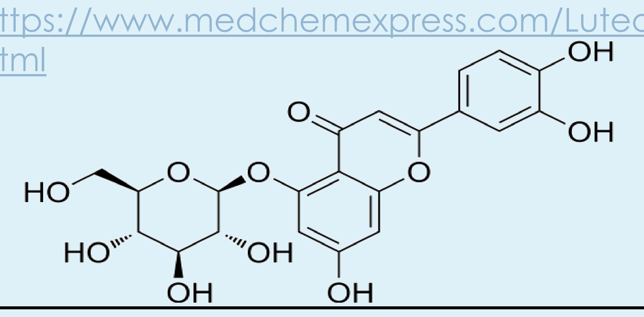


Figure 5: The chemical structure of luteolin https://www.medchemexpress.com/Luteolin.html



Sulforaphane

Sulforaphane arrests the RAF/MEK/ERK pathway, in turn, prohibiting the growth and migration of cancer cells.

Luteolin

Luteolin incudes apoptosis and necrosis, two forms of cell death, in both intrinsic and extrinsic pathways of breast cancer cells.

Purpose & Hypothesis

This project will determine the efficacy of both treatments combined and determine whether they work in synergy with each other.

IV: Concentrations of sulforaphane and luteolin

DV: Measured amount of cell viability

It is expected that these drugs work in synergy with each other because both drugs will be able to target different parts of the ERK pathway and will, therefore, work more effectively when combined. Synergistic treatments are treatments whose combined effects are more efficacious than the expected additive effects.

Methods

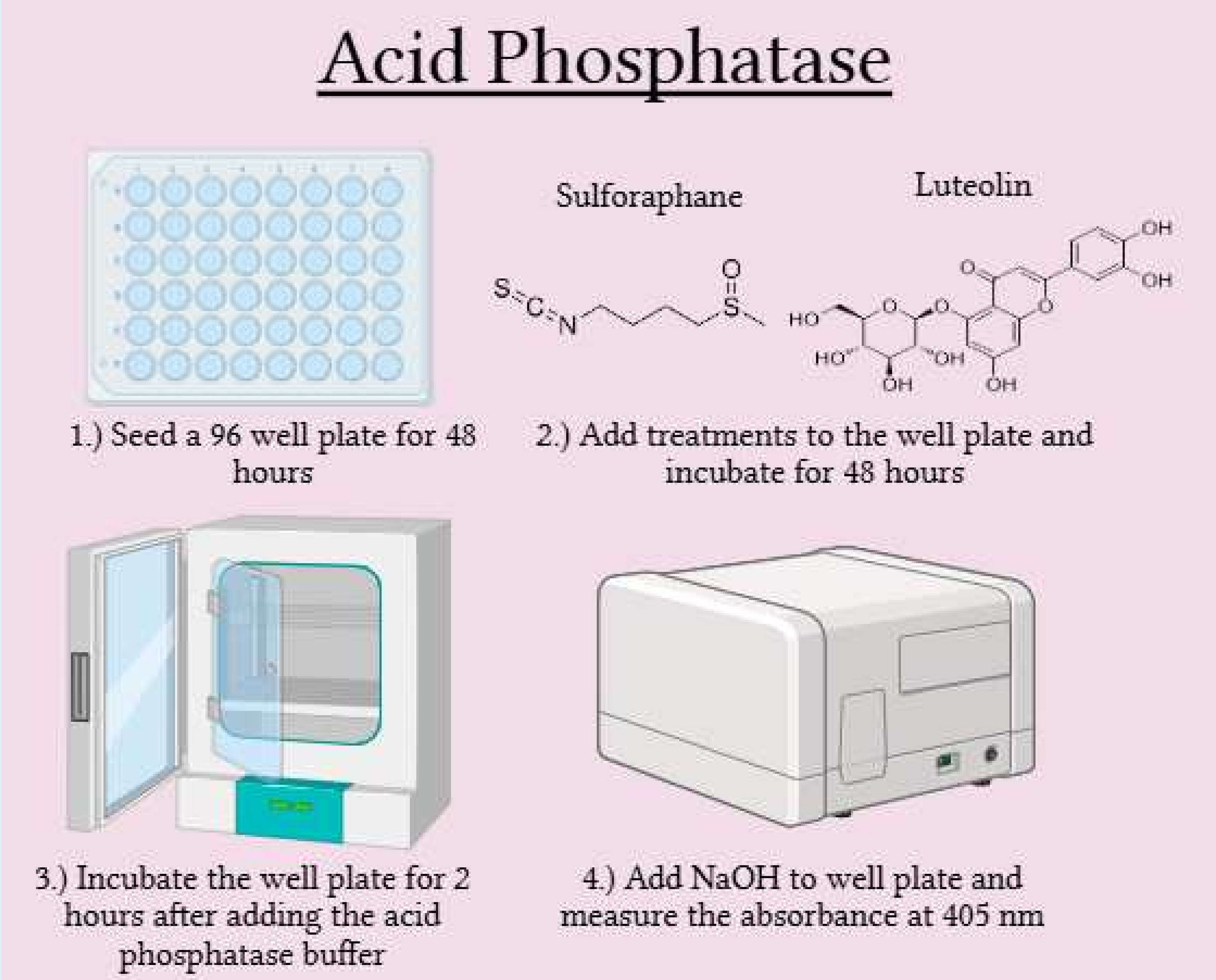


Figure 7: A visual representation of acid phosphatase

Results

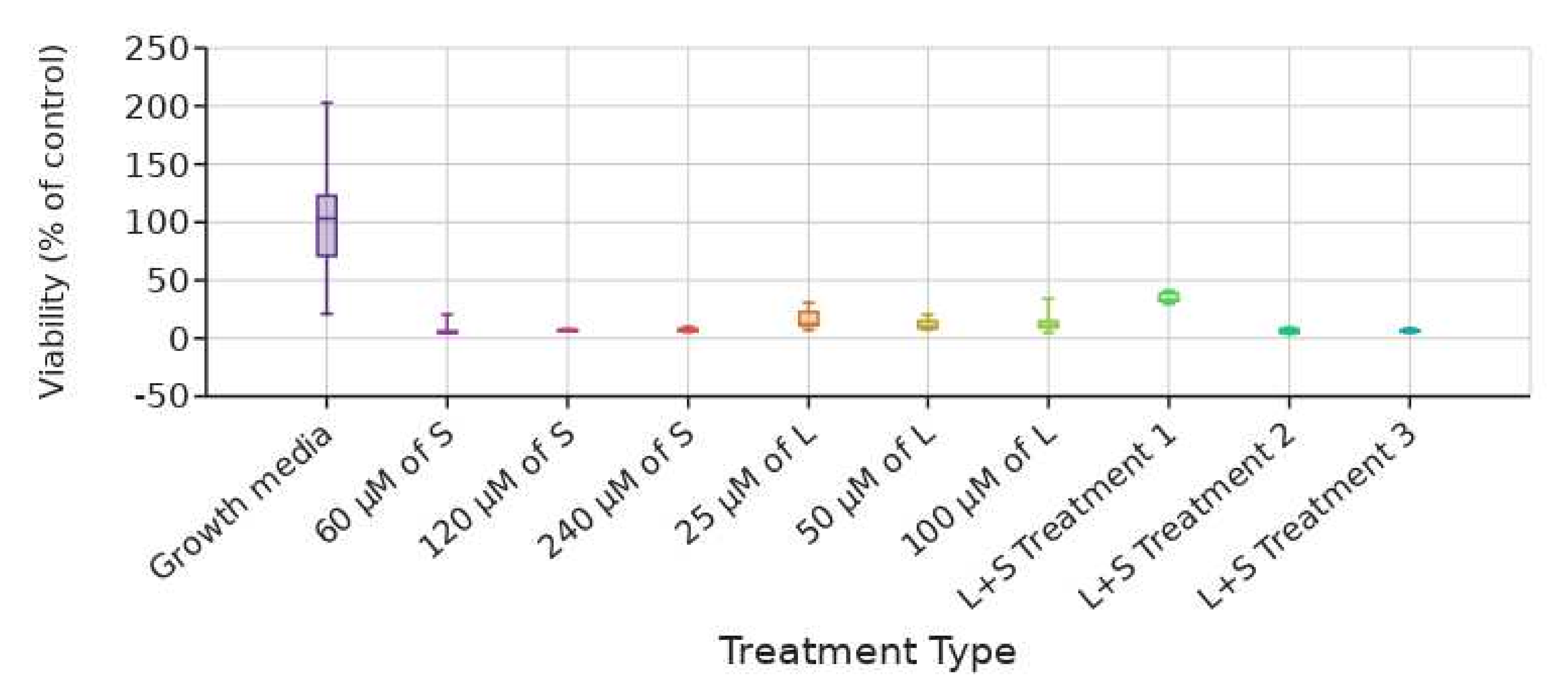


Figure 8: A graph showing the treatment type vs. cell viability

- Growth media
- 120 µM of sulforaphane
- 25 µM of luteolin
- 100 µM of luteolin
- 60 µM of sulforaphane
- 240 µM of sulforaphane
- 50 µM of luteolin
- 60 µM of sulforaphane/ 25 µM of luteolin
- 60 µM of sulforaphane/25 µM of luteolin
- 60 µM of sulforaphane/25 µM of luteolin

Table 1: Data displaying the results of the Kruskal Wallis test

Degrees of Freedom (df)	H-statistic	P-value	Interpretation of P
9	148.25	<0.01	A P-value of <0.01 means extremely strong confidence that the groups are different.

Discussion

Kruskal Wallis results:

- The p-value of less than 0.01 was lower than the critical value of 0.05
- This indicated that the null hypothesis can be rejected meaning that all the treatment groups had extremely different results on the 4T1 cells

Individual and Combined treatment results:

- According to the box and whiskers graph, all the individual treatments resulted in no dose response from the cells
- Because the individual treatments already killed roughly 95% of cells, it was not possible to determine if the two treatments had a synergistic effect when combined
- In order to test synergy in the future, it is necessary to lower the concentrations of the sulforaphane and luteolin treatments

Post Hoc Test:

- Indicated that there were **extreme differences** between the control group and all the treatment groups except for the lowest concentration of the combined treatments which had **little to no indication that it was different** from the control group
- One potential reason as to why this treatment showed these results is that both treatments had been extremely diluted with growth media up to that point, so it is possible that this dilution caused them to be less effective in killing the cells

Future Work

Next year, this project's purpose will be to find treatment concentrations of each sulforaphane and luteolin that are low enough to cause a less intense effect on the cells. This means that these modified concentrations should kill the majority of cells, but not all of them, so that it can be determined if the combined treatments have a greater effect on cell viability than the individual treatments. In addition to acid phosphatase, another assay will be run in order to determine the number of cells that die through the apoptosis pathway specifically. The original plan was to run annexin V which tests for apoptosis in cells, yet this may not be possible because the procedure for annexin V is not designed for adherent cells.

References

Ali, M., Khan, N., Kaleem, N., Ahmad, W., Alharethi, H., Alharbi, B., Alhassan, H., Al-Enazi, M., Razis, A., Modu, B., Calina, D., & Sharifi-Rad J. (2023). Anticancer properties of sulforaphane: current insights at the molecular level. *Front. Oncol.* 13:1168321. doi: 10.3389/fonc.2023.1168321

Guo, Y., Pan, W., Liu, S., Shen, Z., Xu, Y., & Hu, L. (2020). ERK/MAPK signalling pathway and tumorigenesis (Review). Experimental and Therapeutic Medicine, 19, 1997-2007. https://doi.org/10.3892/etm.2020.8454.

Kaboli, P., Khoshkbejari, M., Mohammadi, M., Abiri, A., Mokhtarian, R., Vazifemand, R., Amanollahi, S., Sani, S., Li, M., Zhao, Y., Wu, X., Shen, J., Cho, & C., Xiao, Z. (2020). Targets and mechanisms of sulforaphane derivatives obtained from cruciferous plants with special focus on breast cancer – contradictory effects and future perspectives, Biomedicine & Pharmacotherapy, Volume 121, 109635, ISSN 0753-3322, https://doi.org/10.1016/j.biopha.2019.109635.

Liang, Y., Zhang, H., Song, X., & Yang, Q. (2020). Metastatic heterogeneity of breast cancer: Molecular mechanism and potential therapeutic targets, Seminars in Cancer Biology, Volume 60, Pages 14-27, ISSN 1044-579X, https://doi.org/10.1016/j.semcancer.2019.08.012.

Rocchetti, M. T., Bellanti, F., Zadorozhna, M., Fiocco, D., & Mangieri, D. (2023). Multi-faceted role of luteolin in cancer metastasis: EMT, angiogenesis, ECM degradation and apoptosis. *International Journal of Molecular Sciences*, *24*(10), 8824. https://doi.org/10.3390/ijms24108824.